MIL-C-50980A (AR)
16 March 1981
SUPERSEDING
MIL-C-50980 (AR)
10 January 1972

MILITARY SPECIFICATION

CHARGE, PROPELLING, FOR USE WITH 105MM CARTRIDGES, M393A2, HEP-T; M393A1, TP-T; M416; WP-T; M467, TP-T LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the LAP of a propelling charge designated for use with:

Cartridge, 105MM, HEP-T, M393A2
Cartridge, 105MM, TP-T, M393A1
Cartridge, 105MM, WP-T, M416
Cartridge, 105MM, TP-T, M467

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a aprt of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-46252 - Propellant, M1, For Use With 105MM Cartridges
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8853734 - Charge, Propelling, for use with 105MM Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR/1-199 Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "Code of Federal Regulations, 49 CFR/1-199 (latest revision)."

3. REQUIREMENTS

3.1 Materials.-All materials shall be in accordance with the applicable drawings and specifications.

3.2 Propelling charge.-The propelling charge shall comply with all the requirements specified on drawing (dwg) 8853734 and with the requirements specified herein.

3.3 Propelling charge weight.-The charge weight of M1 propellant shall comply with the weight specified in the loading authorization prescribed by MIL-P-46252 within a tolerance of plus or minus 0.25 ounces.

3.4 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.5 Workmanship

3.5.1 General.-The propelling charge shall be free of dirt, grease and other foreign material.

3.5.2 Marking.-Each propelling charge shall be marked in accordance with dwg. 8853734.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

Part Description	Drawing	Quantity
Propelling Charge	8853734	10
Propellant Bag	8853733	10
Lead Foil	8853731	10

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I FIRST ARTICLE TESTS

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
	Charge, Propelling, For Use With 105MM Cartridges M393A2, HEP-T; M393A1, TP-T, M416, WP-T; M467, TP-T	1	1	See below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Foil, Lead (Dwg. 8853731) Examination for defects	10		3.1	4.4.2.1
	Bag, Propellant (Dwg. 8853733) Examination for defects	10		3.1	4.4.2.2
	Propelling Charge (Dwg. 8853743) Examination for defects	10		3.1	4.4.2.4

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of propelling charges shall contain no more than one lot of M1 propellant.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONF ORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50980A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Foil, Lead				8853731
					NEXT HIGHER ASSEMBLY
					8853733
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Length		0.65%	3.1	Gage
202	Width		0.65%	3.1	Gage
203	Evidence of poor workmanship		0.65%	3.5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50980A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8853733
4.4.2.2	Bag, Propellant				NEXT HIGHER ASSEMBLY 8853734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Number of pockets incorrect		0.40%	3.1	Visual
102	Lead foil missing		0.40%	3.1	Visual
103	Bag torn or otherwise damaged		0.40%	3.1	Visual
104	Safety stitch missing		0.40%	3.1	Visual
Minor					
201	Width of bag		0.65%	3.1	Gage
202	Length of bag		0.65%	3.1	Gage
203	Width of pocket		0.65%	3.1	Gage
204	Any pocket not open at one end		0.65%	3.1	Visual
205	Stitching improper		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.5.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8853734
4.4.2.3	Propellant, Prior to Assembly in Propelling Charge				NEXT HIGHER ASSEMBLY 8853734
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Propellant type incorrect			3.3	Visual
Major					
101	Charge weight incorrect	a		3.3	Balance
Minor	None defined				

NOTES:

A. See 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50980A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Propelling Charge				8853734
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any opening in bag large enough to permit escape of propellant grains		100%	3.2	Visual
102	Any opening in bag, but not large enough to let propellant grains to escape		0.40%	3.2	Visual
103	Stitch skipped or broken		0.40%	3.2	Visual
104	Safety Stitch missing		0.40%	3.2	Visual
105	Lead foil damaged		0.40%	3.2	Visual
Minor					
201	Seam less than 3/16 inch from outer edge		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.5.1	Visual
203	Marking illegible, incorrect or missing		0.65%	3.5.2	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Propelling charge weight, Major Defect.-The propellant shall be weighed and checkweighed 100% using a calibrated and approved scale. The checkweighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirements prescribed herein shall be classed defective and removed from the lot.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.2 herein and Section 6 of MIL-A-48078.

4.5 Methods of inspection

4.5.1 Precaution.-This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

5. Packaging

5.1 Packing.-Propellant charge assemblies shall be packed in accordance with the Code of Federal Regulations 49 CFR/1-199.

5.2 Marking.-Exterior containers shall be marked in accordance with the manufacturer's normal practice and in conformance with the Department of Transportation requirements for shipment of propellant. The DOT marking shall be "PROPELLANT EXPLOSIVE - CLASS A".

6. NOTES

6.1 Intended use.-The propelling charges covered by this specification are intended for use in the following 105MM cartridges: HEP-T, M393A2; WP-T, M416; and TP-T, M393A1 and M467.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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6.4 Supersession data.-This specification supersedes those paragraphs pertaining to the Loading, Assembling and Packing of propelling charges contained in:

MIL-C-60039 - Cartridge, 105MM, HEP-T, M393A2, LAP
MIL-C-46956 - Cartridge, 105MM, WP-T, M416, LAP
MIL-C-46665 - Cartridge, 105MM, TP-T, M467, LAP

6.5 Changes from previous issue.-Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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